

CHAHEL PAATUR

Physical AI · Embedded Systems · Autonomous Robotics

Tracy, CA · [Click Below For Links](#)

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EDUCATION

John C. Kimball High School — *Tracy, CA*

GPA 3.83 (Unweighted) · 4.4 (Weighted)

Washington High School · Prior Coursework

TECHNICAL SKILLS

Languages Python, C/C++, Java, JavaScript, TypeScript, SQL

AI / ML Reinforcement Learning (PPO), Neural Networks, PyTorch, Computer Vision

Embedded Raspberry Pi, ESP32, Embedded Linux, Real-Time Systems

Robotics FPV Drones, RF Systems, Sensor Fusion, Flight Control

Tools React, Node.js, REST APIs, Git, Fusion 360, 3D Printing

Research Statistics, MATLAB, Quantitative Modeling

PROJECTS

AETHER — *Adaptive Embodied Task Hierarchy for Robotics*

- Built a physical AI system that translates natural language into executable robotic actions.
- Designed full pipeline: perception → planning → execution → feedback → adaptation.
- Integrated Vision-Language-Action (VLA) frameworks for cross-platform deployment across drones, rovers, and humanoids.

Autonomous AI Drone — *FPV + Embedded Compute*

- Engineered custom drone with SpeedyBee F7 flight controller and onboard Raspberry Pi compute.
- Integrated AI decision-making with low-level motor control; bridged autonomy algorithms with real-world flight.
- Solved latency, power distribution, and signal interference challenges in production hardware.

SAURON — *RF-Based Search & Rescue System* 2nd Overall + Palantir Award

- Built distributed RF localization system using ESP32 nodes with RSSI-based triangulation.
- Implemented drone verification pipeline robust to signal noise, obstruction, and interference.

Spacecraft Fault Detection AI — *FDIR — Aerospace Safety Systems*

- Developed hybrid DRL + rule-based system for safety-critical fault detection.
- Implemented PPO agent with safety constraints; simulated 100+ failure scenarios.

Atlas — *AR-Assisted Engineering Workstation*

- Designing desk-mounted physical AI assistant with camera, projection, and laser guidance.
- Detects wiring and components; overlays real-time guidance via voice activation.

Ionic Thruster — *Experimental Propulsion System*

- Designed and tested small-scale electrostatic propulsion; explored ion acceleration and thrust efficiency.

Gaussian — *Founder — Custom Programming Language*

- Built complete compiler architecture; reached 200–300+ downloads with active usage.

NueroLang — *Neural Network Framework*

- Developed framework enabling neural networks in ~12 lines of code with simplified API abstractions.

RESEARCH & PUBLICATIONS

- Self-Modifying Program Synthesis via Online Library Evolution — arXiv (2026)
- Microwave Technology in Aeronautical Applications (2025)
- DRL vs Rule-Based Systems for Spacecraft FDIR (2025)
- Quantum vs Classical vs Hybrid Computing Paradigms (2025)

Impact: 200–400+ total reads/downloads · Recognized in the IEEE aerospace context

PATENTS

- Non-Provisional Patent — AI-Based Spacecraft Fault Detection System (Ref. 63/842,839)
- Provisional Patent (in progress) — Eye-Tracking + Gesture-Based AR Interface for Assistive Tech
- Regeneron Science Talent Search — Aerospace AI Fault Detection (FDIR) submission

RESEARCH EXPERIENCE

UC Irvine — GATI Program — *Research Intern*

- Conducted applied AI and engineering research; developed experimental and analytical methodologies.

Independent & Collaborative Research — *AI · Aerospace · Physical Systems*

- Collaborated with peers across advanced academic networks (including Princeton-level circles).
- Built a technical research archive spanning multiple domains.

LEADERSHIP & ACHIEVEMENTS

PII Research Network — *International Vice President*

- Expanded organization to 20+ schools globally; built a community of 1,000+ student researchers.
- Led mentorship and engineering project pipelines.

Science Olympiad — *Regional Champion*

- 1st Place Regional — Chemistry event; contributed to multi-event team placements.

Additional: Built platforms reaching 10K+ users globally · Varsity tennis athlete